

ARNAV RAJ

Full-Stack Developer | AI & RAG Systems Engineer

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PROFESSIONAL SUMMARY

Full-Stack Developer with hands-on experience shipping production web applications using Next.js, React, Node.js, Prisma, and PostgreSQL, plus applied experience building Retrieval-Augmented Generation (RAG) systems with LangChain, FastAPI, and ChromaDB. Founded and built CareerSathi, a live AI-powered college admissions platform aggregating verified data (AISHE, NAAC, NIRF) for 4,760+ colleges across India. Experienced in AI model evaluation and code-quality review through freelance work training large language models. Strong foundation in Data Structures & Algorithms, full-stack architecture, and iterative, metrics-driven development.

TECHNICAL SKILLS

- Languages:** JavaScript, TypeScript, Python, C/C++, SQL, HTML/CSS
- Frontend:** React.js, Next.js, Tailwind CSS, Responsive UI Design
- Backend & Databases:** Node.js, FastAPI, Prisma ORM, PostgreSQL, Supabase, REST APIs
- AI / ML:** Retrieval-Augmented Generation (RAG), LangChain, ChromaDB (Vector Databases), Prompt Engineering, LLM Evaluation & Hallucination Testing, Conversational AI / Chatbot Development
- Cloud & Tools:** AWS (EC2, S3, RDS, Lambda, IAM, VPC), Git, GitHub, Vercel, Render, VS Code
- Core CS:** Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

EXPERIENCE

Founder & Full-Stack Developer | CareerSathi (CourseAdmission)

Oct 2025 – Present

Remote | careersathhi.com

- Built and shipped a production-ready AI-powered college admissions platform serving students across India, covering 4,760+ colleges, using Next.js 16, React.js, Prisma ORM, and PostgreSQL.
- Designed a B2B2C data model aggregating verified data from AISHE, NAAC, and NIRF; built a Python pipeline to automate downloading, normalizing, and merging institutional datasets.
- Developed “Q&A GPT,” an AI-powered conversational chatbot for the admissions help desk, delivering fast, accurate responses to student queries.
- Built an admin dashboard with lead capture, college comparison/filtering, and Excel export for partner colleges.

AI/RAG Systems Developer | Personal Project — Shiksha RAG

Sep 2026

github.com/arnav20719/shiksha-rag

- Designed and built a Retrieval-Augmented Generation pipeline (LangChain, FastAPI, ChromaDB) to answer college-admissions queries using chunked, metadata-tagged institutional data.
- Built an iterative evaluation framework, scaling from 5 to 40 test cases and improving measured retrieval accuracy from 76.7% to 100% through hybrid retrieval tuning and adversarial hallucination testing.
- Diagnosed and resolved a production deployment constraint on Render by redesigning the vector store to run in-memory on the free tier.
- Paired the backend with a Next.js/TypeScript frontend (“Shiksha Chat”) — [Live Demo](#)

Software Engineer – AI Training (Code Quality & Debugging) | Outlier (Freelance)

Oct 2023 – Jul 2025

Remote

- Reviewed, debugged, and rated AI-generated code across multiple languages to improve large language model training data quality.
- Applied Data Structures & Algorithms knowledge to evaluate correctness, efficiency, and edge-case handling in model-generated solutions.

EDUCATION

B.Tech, Electronics and Communication Engineering

Aug 2021 – Aug 2025

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science & Technology, Chennai | CGPA: 8.5

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, OOP, Database Management Systems, Computer Networks, IoT

CERTIFICATIONS

- [Prompt Engineering for Developers: Master AI Prompting & Automation](#) — Udemy, Mar 2026
- [The Complete Full-Stack Web Development Bootcamp \(React, Node, PostgreSQL\)](#) — Udemy, Jan 2026
- [The Complete Data Structures with Python](#) — Udemy
- AWS Cloud Virtual Internship — AICTE, May–Jun 2023 | [Coding Competition Participation Certificate](#) — Unstop